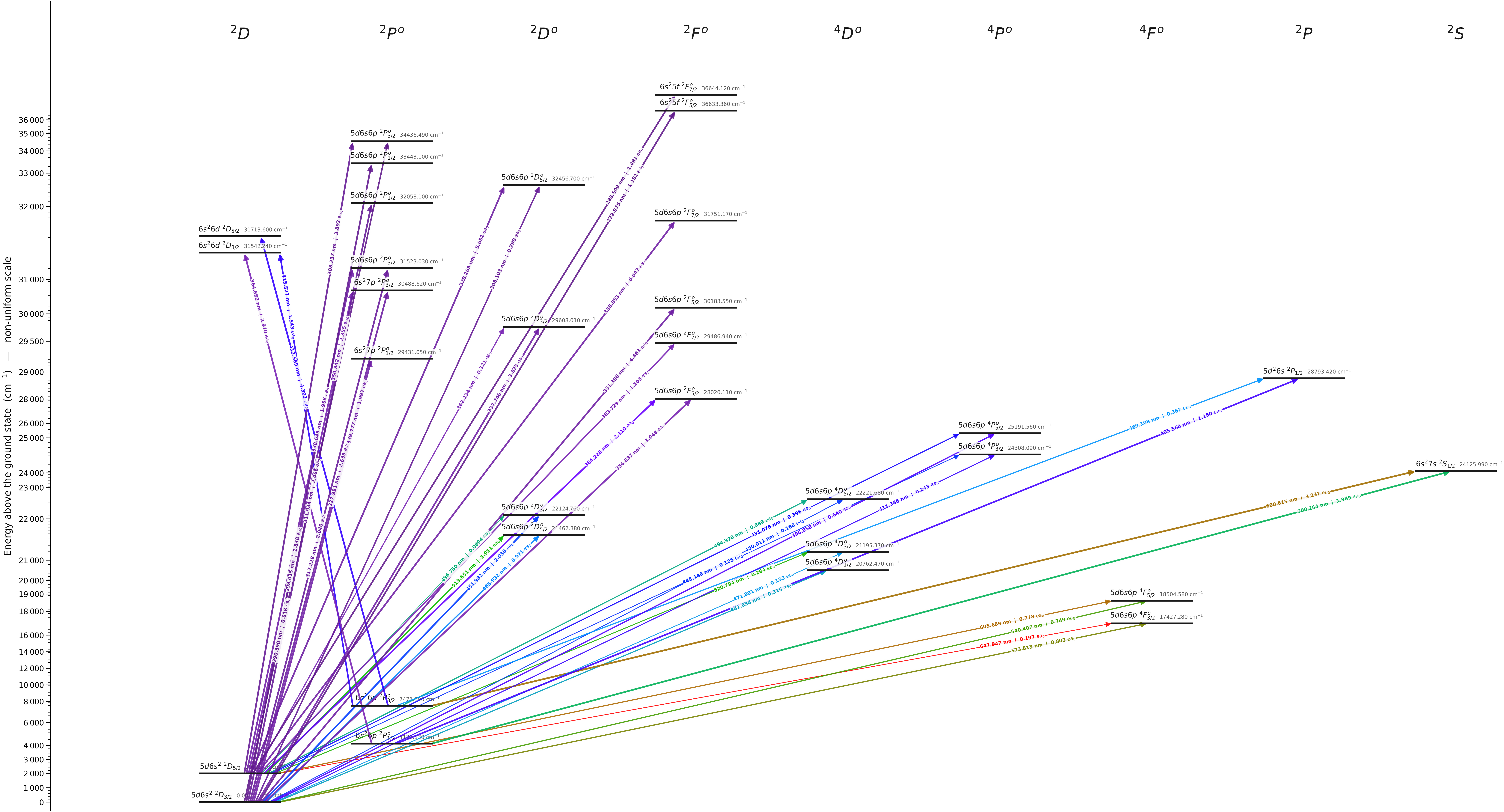
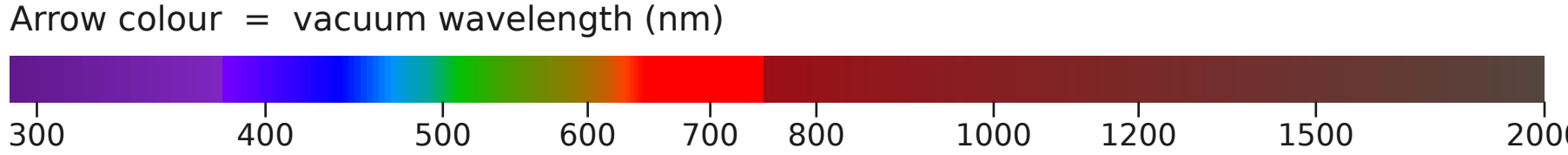


Lutetium

Lu (Lu I). Strongest classified transitions below 2 μm (NIST): 31 levels, 44 lines.

Lu⁺ ionisation limit 43762.60 cm⁻¹ = 5.42587 eV (228.506 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle U \| er \| J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d6s ² ² D _{3/2} - 5d6s6p ² P _{1/2}	299.0153	298.9281	1002.5989	1.838	1.280e+08	-	-	-
5d6s ² ² D _{5/2} - 5d6s6p ² P _{3/2}	308.2370	308.1475	972.6038	3.892	2.620e+08	-	-	-
5d6s ² ² D _{3/2} - 5d6s6p ² P _{3/2}	311.9336	311.8432	961.0777	2.466	2.030e+08	-	-	-
5d6s ² ² D _{3/2} - 5d6s6p ² P _{3/2}	317.2284	317.1366	945.0367	2.040	6.600e+07	-	-	-
5d6s ² ² D _{3/2} - 6s ² 7p ² P _{3/2}	327.9912	327.8967	914.0258	2.639	1.000e+08	-	-	-
5d6s ² ² D _{5/2} - 5d6s6p ² D _{3/2}	328.2694	328.1749	913.2512	5.652	3.050e+08	-	-	-
5d6s ² ² D _{3/2} - 5d6s6p ² F _{5/2}	331.3063	331.2110	904.8801	4.463	1.850e+08	-	-	-
5d6s ² ² D _{5/2} - 5d6s6p ² F _{7/2}	336.0526	335.9560	892.0999	6.047	2.440e+08	-	-	-
5d6s ² ² D _{3/2} - 5d6s6p ² D _{5/2}	337.7464	337.6495	887.6258	3.575	1.680e+08	-	-	-
5d6s ² ² D _{5/2} - 5d6s6p ² P _{3/2}	338.6489	338.5517	885.2604	1.958	5.000e+07	-	-	-
5d6s ² ² D _{3/2} - 6s ² 7p ² P _{1/2}	339.7772	339.6797	882.3207	1.997	1.030e+08	-	-	-
5d6s ² ² D _{5/2} - 6s ² 7p ² P _{3/2}	350.9425	350.8421	854.2496	2.355	6.500e+07	-	-	-
5d6s ² ² D _{3/2} - 5d6s6p ² F _{5/2}	356.8865	356.7847	840.0218	3.048	6.900e+07	-	-	-
6s ² 6p ² P _{1/2} - 6s ² 6d ² D _{3/2}	364.8821	364.7782	821.6145	2.970	9.200e+07	-	-	-
6s ² 6p ² P _{3/2} - 6s ² 6d ² D _{5/2}	412.5887	412.4724	726.6133	4.302	8.900e+07	-	-	-
6s ² 6p ² P _{3/2} - 6s ² 7s ² S _{1/2}	600.6150	600.4487	499.1424	3.237	4.900e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).